AI-Style in Transformer Residual Streams: Representation-Causation Dissociation and the Style-Coherence Tradeoff

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Abstract

Large language models (LLMs) produce outputs with a recognizable stylistic signature that humans readily identify as AI-generated. Prior work has demonstrated that transformer language models encode the property of "sounding like AI" as a near-linear direction in the residual stream, extractable via Contrastive Activation Analysis (CAA); however, these findings are substantially confounded by text length [Hypogenic-AI, 2024]. In this work, we investigate whether this direction is genuine beyond verbosity and, critically, whether the layer from which it is most cleanly extracted is also causally operative during generation. Using CAA on a length-matched version of the HC3 dataset, we extract a genuine AI-style direction classifiable at 96.5-98.0% accuracy on both QWEN2.5-3B and QWEN2.5-3B-INSTRUCT. We then find a sharp dissociation between representation and causation: the layer that most linearly encodes the extracted direction is causally inert, while a distinct region in the middle of the network drives stylistic generation. We additionally propose and evaluate a detector-weighted variant of CAA. For both CAA methods, we find that ablation consistently incurs a writing coherence cost, and a Pareto analysis of AI-style ablation versus writing coherence retention reveals that while both CAA methods achieve comparable ablation, the GPTZero-weighted direction yields meaningfully better coherence preservation in QWEN2.5-3B-INSTRUCT. More broadly, our work showcases how an understanding of model internals can be leveraged to develop more robust AI-detection and AI-style intervention techniques.

1 Introduction

LLM outputs have a distinctive stylistic signature that humans readily recognize. They often have minimal hedged phrasing, a more formal register, and bullet-pointed structuring. AI text detection systems increasingly inform policy decisions in academic integrity, content moderation, and synthetic media governance, yet the mechanistic encoding of AI-style within transformer models remains poorly understood. If the model itself represents AI-style as a structured internal direction, this opens principled paths to both more robust detection and more natural generation. Inspired by the rapid progress of mechanistic interpretability, this works leverages the internal representations of LLMs to better understand how AI-style is encoded.

It is widely hypothesized that LLMs represent high-level semantic features, or concepts, as linear directions in residual stream activations [Park et al., 2023]. A growing body of work has extracted causal linear directions for specific behaviors: truthfulness [Marks and Tegmark, 2024], sentiment [Konen et al., 2024], refusal Arditi et al. [2024], and sycophancy [Panickssery et al., 2024]. Moreover, these feature directions have been shown to be effective casual mediators of behavior, enabling fine-grained steering of model outputs with minimal capability cost. Thus, this work aims to provide insight on whether "sounding like AI" is linearly encoded in the residual stream, and whether that encoding is casually operative during generation.

Prior work has demonstrated that transformer language models encode the property of "sounding like AI" as a near-linear direction in the residual stream, classifying AI versus human text with 97.5% accuracy via CAA on the HC3 dataset. However, these findings are substantially confounded by text length with a cosine similarity of 0.93 between the "AI direction" and a length-based direction [Hypogenic-AI, 2024].

We make the following **contributions:**

- We construct a length-matched variant of HC3 and extract a genuine AI-style direction achieving 96.5–98.0% accuracy on both QWEN2.5-3B and QWEN2.5-3B-INSTRUCT, with the verbosity confound explicitly eliminated (length cosine similarity drops from 0.93 to -0.065).
- Via a systematic sliding-window layer sweep ($k \in \{1, 5\}$ across all 36 layers), we identify a representation-causation dissociation: the best-classifying layer (28-29) is causally inert during generation, while the middle layers constitute the true causal generation region.
- We show that ablating the AI-style direction almost always incurs a writing coherence cost, with magnitude varying by model, intervention type, and writing category. We introduce a Pareto frontier analysis jointly measuring AI-style ablation and writing coherence retention, identifying the optimal intervention window (layers 18-25, $k = 5$) and showing that while detector-weighted CAA does not improve upon standard CAA in ablation power, DW-GPTZero achieves meaningfully better coherence preservation in QWEN2.5-3B-INSTRUCT ($\sim$17% relative improvement).
- Across all ablation experiments, AI-style ablation is largely disjoint from general factual and commonsense capabilities as MMLU and HellaSwag remain near baseline, while TruthfulQA and WikiText-2 perplexity reveal a base-instruct divergence consistent with RLHF geometrically separating truthfulness from the style direction (Appendix K).

2 Methodology

2.1 Dataset Construction

We use HC3 [Guo et al., 2023], a corpus of about 40K paired human and ChatGPT responses drawn primarily from the ELI5 subreddit and several question-answering sources. The original dataset exhibits a systematic length asymmetry: AI responses average approximately 914 characters versus 446 for human responses with standard deviations of 340 and 279 characters respectively [Hu et al., 2024]. An AI-style CAA direction extracted from HC3 will therefore be significantly confounded by verbosity.

To control for the verbosity confound, we construct a *length-matched* variant using a sentence-boundary-aware truncation algorithm. For each (human, AI) pair, we identify the shorter response and truncate the longer to the same character count, snapping to the nearest sentence boundary to avoid mid-sentence cuts. Pairs with a length ratio exceeding 5.0 or a post-truncation length below 50 characters are discarded. The resulting dataset has 30,758 (human, AI) pairs with mean lengths of approximately 519 characters for human responses and 468 characters for AI responses with standard deviations of 277 and 295 respectively. We use a 200/50/100 train/validation/test split matching prior work [Hypogenic-AI, 2024] on this dataset for direct comparability.

2.2 Models

We study two variants from the Qwen2.5 family [Qwen et al., 2025]. QWEN2.5-3B is a 3B-parameter decoder-only transformer with 36 layers and a hidden dimension of 2048, trained on a large multilingual pretraining corpus. QWEN2.5-3B-INSTRUCT is the corresponding instruction-tuned variant, fine-tuned via reinforcement learning from human feedback (RLHF) for helpfulness and safety. The base model serves as the primary subject for direction extraction, following prior work. The instruct model allows us to examine how RLHF-induced alignment reshapes the geometry of the style direction. All experiments are run on a single NVIDIA A10G GPU via Modal cloud compute.

2.3 Direction Extraction

To extract the AI-style direction, we use two forms of Contrastive Activation Analysis: (1) Standard and (2) Detector-Weighted.

2.3.1 Standard CAA

We follow the definition of CAA as defined in [Panickssery et al., 2024]. For each text in the training split, we perform a forward pass and record the last-token residual stream activation $\mathbf{h}^{(l)} \in \mathbb{R}^{2048}$ at every layer $l \in \{0, \ldots, 35\}$ where layer 0 is defined as the embedding layer. The CAA direction at layer l is the difference of class means:

$$\mathbf{d}_{\text{CAA}}^{(l)} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{h}_{\text{AI},i}^{(l)} - \frac{1}{N} \sum_{i=1}^{N} \mathbf{h}_{\text{human},i}^{(l)} \tag{1}$$

where N is the number of pairs. We normalize each $\mathbf{d}_{\text{CAA}}^{(l)}$ to unit length and select the best layer l^* by maximizing linear classification accuracy on the validation set. The resulting unit vector $\hat{\mathbf{d}} = \frac{\mathbf{d}^{(l^*)}}{\|\mathbf{d}^{(l^*)}\|}$ is used for all downstream ablation experiments unless otherwise specified.

2.3.2 Detector-Weighted CAA

Standard CAA treats all training examples equally, so the direction reflects an average contrast that may be influenced by non-stylistic distributional features. However, AI detectors implicitly encode a continuous notion of how "AI-sounding" a piece of text is, and a text that confidently fools an AI detector may carry less of the relevant stylistic signal than one flagged at 100% confidence. We therefore propose a detector-weighted variant of CAA:

$$\hat{\mathbf{d}}_{\text{weighted}}^{(l)} = \frac{\sum_i s_i \, \mathbf{h}_{\text{AI},i}^{(l)}}{\sum_i s_i} - \frac{\sum_i (1 - s_i) \, \mathbf{h}_{\text{human},i}^{(l)}}{\sum_i (1 - s_i)} \tag{2}$$

where $s_i \in [0, 1]$ is the detector's AI-probability for the i-th training text. Detector-weighted CAA re-weights examples by the confidence of a pre-trained AI detector, so texts that the detector is more confident are AI-generated contribute more to the AI-side mean (have a higher weight), and texts the detector considers least AI-like contribute less. In other words, highly AI-like examples dominate the direction and ambiguous examples will contribute less.

We use three detectors with distinct decision boundaries for detector-weighted CAA. **(1) RoBERTa** [Liu et al., 2019] (`openai-community/roberta-base-openai-detector`) is a fine-tuned sequence classifier operating primarily on surface lexical distributional features (vocabulary register, punctuation patterns, n-gram statistics). **(2) RADAR** [Hu et al., 2023] is a DeBERTa-v3 classifier trained in an adversarial loop against a Vicuna-7B paraphraser, making it robust to surface style-transfer and potentially sensitive to deeper syntactic and semantic patterns. **(3) GPTZero** [Tian and Cui, 2023] is a commercial detector combining per-sentence perplexity, burstiness, and a trained classifier, with its decision boundary most sensitive to local predictability.

2.4 Ablation Hook

To causally test the role of a layer or layer window, we register a persistent forward hook that orthogonally projects out the AI-style direction from the residual stream at every forward pass during generation:

$$\mathbf{h}_{\text{ablated}} = \mathbf{h} - (\mathbf{h} \cdot \hat{\mathbf{d}}) \, \hat{\mathbf{d}} \tag{3}$$

We evaluate three ablation conditions: (1) **Baseline**: no hook applied, (2) **Best-layer ablation**: hook at the single best-classification layer l^* only, (3) **All-layer ablation**: hook at all 36 layers simultaneously. In all ablation conditions, the *same* unit direction $\hat{\mathbf{d}} = \hat{\mathbf{d}}^{(l^*)}$ extracted at the best-classifying layer l^* is projected out at every hooked layer. The hook does not apply a layer-specific direction as it removes the single globally extracted direction from whichever layers fall within the ablation window.

2.5 Layer-Window Sweep

For each window of $k \in \{1, 5\}$ consecutive layers with center position $c \in \{0, \dots, 35\}$, we apply the ablation hook only within that window and generate outputs. Sliding c across all 36 positions yields a causal profile of the style direction across the full network depth. The two choices of k correspond to qualitatively different interventions: $k = 1$ provides maximal localization, revealing fine-grained causal peaks, while $k = 5$ offers robustness against imprecise layer targeting and represents a larger intervention size.

2.6 Evaluation

AI detection on HC3 test set. We report RoBERTa AI-probability scores [0,1] on held-out HC3 test responses. Human and AI reference scores on the HC3 test set are 0.036 and 0.870, respectively. For detector-weighted runs, we additionally report scores from the corresponding detector.

LLM-as-judge for writing prompts. We use Claude Sonnet 4.6 as an automated judge to rate outputs on AI-likeness (does this read as AI-generated?), writing quality (coherence, fluency, structure), and factual quality, each on a 1-7 scale. Ratings are collected on 20 MT-Bench-style writing prompts spanning five categories: creative writing, persuasive/argumentative, expository, narrative/reflective, and professional/functional. We report mean scores with standard deviations over $n = 20$ samples for writing prompt outputs and over $n = 25$ samples for HC3 prompt outputs. LLM-judge prompts can be found in Appendix H.

Capability benchmarks. We evaluate on four benchmarks: (1) MMLU (500 questions across 10 subjects, 5-shot), (2) HellaSwag (200 commonsense completion questions, 0-shot log-probability scoring), (3) TruthfulQA MC1 (200 truthfulness questions, 0-shot), and (4) WikiText-2 perplexity (200 held-out sentences).

Pareto frontier analysis. To formally characterize the style–coherence tradeoff across all ablation conditions and runs, we define a Pareto frontier where each experimental condition is represented as a point with $x = \mathrm{AI}_0 - \mathrm{AI}_i$ (AI-style ablation, higher is better) and $y = C_i / C_0$ (coherence retention, higher is better), where AI_0 and C_0 are the unablated baseline scores. A condition is Pareto-optimal if no other condition achieves both greater ablation and higher coherence retention simultaneously.

Logit-lens vocabulary projection. We project the AI-style direction $\hat{\mathbf{d}}$ onto the model's unembedding matrix $W_U \in \mathbb{R}^{V \times d}$ to produce a vocabulary-space signature $\mathbf{v} = W_U \hat{\mathbf{d}}$. Tokens with the largest positive entries in $\mathbf{v}$ are those whose generation probability would increase most if the direction were added to a residual stream hidden state; tokens with the most negative entries form the human-style vocabulary signature.

3 Results

3.1 The AI-style direction is linearly encoded and genuine

We first verify that a genuine AI-style direction exists in the residual stream independently of text length via standard CAA. Table 1 summarizes classification accuracy, AUC, and length cosine similarity across all four conditions: (original, length-matched HC3 datasets) × (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT).

Table 1: Summary of extracted AI-style direction via standard CAA across four conditions. "Length cos. sim." is the cosine similarity between the extracted direction and a pure text-length direction.

Condition	Best Layer	Test Acc. (%)	Test AUC	Length Cos. Sim.
Original (prior Hypogenic-AI work)	21	97.5	0.999	0.930
(1) QWEN2.5-3B (original HC3)	13	97.0	0.997	0.938
(2) QWEN2.5-3B-INSTRUCT (original HC3)	8	96.5	0.995	0.922
(3) QWEN2.5-3B (length-matched)	29	96.5	0.998	−0.065
(4) QWEN2.5-3B-INSTRUCT (length-matched)	29	98.0	0.998	−0.151

Length confound on original HC3. On the original HC3 dataset, the extracted CAA direction has cosine similarity 0.93 with a pure text-length direction, indicating that most of the classification signal is attributable to verbosity rather than style. Appendix A illustrates this confound in detail: projection distributions show clean separation between human and AI activations, but the separation tracks almost perfectly with response length, and the length direction alone achieves near-identical accuracy to the full AI direction.

Genuine style signal on length-matched HC3. After length-matching, the cosine similarity between the extracted direction and the length direction collapses to -0.065 for QWEN2.5-3B and -0.151 for QWEN2.5-3B-INSTRUCT which are near zero, confirming that verbosity has been removed as a confounding signal. Critically, classification accuracy remains at 96.5% and 98.0% respectively, demonstrating that a genuine stylistic signal independent of response length is linearly encoded in the residual stream. Appendix B shows the corresponding analyses: the length-vs-projection scatter shows no relationship between length and the extracted direction, the per-layer classification accuracy plot shows the length-only direction collapsing to below-chance, and PCA continues to reveal clean cluster structure across layers.

Two additional observations are worth noting. First, the best-classifying layer shifts from layer 13-21 on original HC3 to layer 29 on length-matched data for both models, suggesting that pure stylistic information is encoded deeper in the network once the early, surface-level length signal is removed. Second, QWEN2.5-3B-INSTRUCT achieves slightly higher accuracy than QWEN2.5-3B on length-matched data (98.0% vs. 96.5%), consistent with RLHF strengthening the stylistic signal independently of verbosity. All subsequent experiments in this paper use the length-matched HC3 dataset unless otherwise specified.

3.2 Representation-causation dissociation through layer sweep

Motivating observation. Initial ablation experiments revealed a striking result: ablating the best-classifying layer ($l^* = 29$) during generation produces near-zero change in measured RoBERTa AI-likeness. Table 2 shows RoBERTa HC3 AI scores across initial ablation conditions for both models. Single-layer ablation at the best-classifying layer leaves the AI score essentially unchanged relative to baseline, with a $+0.031$ and $+0.006$ change for QWEN2.5-3B and QWEN2.5-3B-INSTRUCT respectively, while all-layer ablation produces a substantial reduction. This dissociation between the layer that most linearly *encodes* the direction and the layers that causally *generate* AI style motivates the systematic sweep below.

Table 2: RoBERTa HC3 AI-probability scores (mean over 100 test outputs) for standard CAA on both models. Human reference = 0.036; AI reference = 0.870. Single-layer ablation at the best-classifying layer ($l^* = 29$) has essentially no effect, while all-layer ablation produces a substantial reduction.

Model	Baseline	Best-layer ablation	All-layer ablation
QWEN2.5-3B	0.840	0.871 (+0.031)	0.651 (-0.189)
QWEN2.5-3B-INSTRUCT	0.656	0.662 (+0.006)	0.196 (-0.460)

Layer sweep results. To identify where in the network AI style is causally generated, we conduct a systematic sliding-window sweep. For each window center $c \in \{0, \dots, 35\}$ and window size $k \in \{1, 5\}$, we apply the ablation hook only within that window and measure both RoBERTa HC3 AI score and LLM-judge AI-likeness on 25 held-out HC3 prompt responses and 20 writing prompts respectively. Full sweep results are shown in Appendix C.

The sweep reveals two key patterns. First, ablating the best-classifying layer (28-29) produces near-zero change in both HC3 AI score and LLM-judge AI-likeness across all conditions, confirming that the representation readout layer is causally inert. Second, a causal generation region consistently emerges in the middle of the network. For QWEN2.5-3B, ablating around layers 15-19 ($k = 1$) reduces LLM-judge AI-likeness from a baseline of $\sim$6.2 to $\sim$4.6 and the RoBERTa HC3 AI score by approximately 37% relative to baseline. For QWEN2.5-3B-INSTRUCT, the causal peak shifts slightly later. Increasing the window size from $k = 1$ to $k = 5$ smooths the causal profile and amplifies the magnitude of AI-likeness reductions, consistent with the style direction being distributed across several consecutive layers rather than sharply localized at a single point.

Interpretation. One explanation for the seen representation-causation dissociation is that the final layers are a downstream reflector of style already committed by middle layers. Linear probing identifies where information is *encoded* most cleanly in the residual stream; it does not identify where that information *causally drives* behavior during generation. This distinction matters for mechanistic interpretability more broadly as high probing accuracy at a given layer is not sufficient evidence that the layer is a causal locus of the probed property.

3.3 Detector-Weighted CAA: Direction Geometry and Ablation Effectiveness

Standard CAA weights all training examples equally, so the extracted direction reflects an average contrast that may be influenced by non-stylistic distributional features. Detector-weighted CAA re-weights by detector confidence so that prototypically AI-like examples dominate the direction. In this section, we analyze ablation effectiveness across detectors with distinct decision boundaries, using both HC3 AI-probability scores and LLM-judge AI-likeness as evaluation metrics. The style–coherence tradeoff for both standard and detector-weighted CAA is examined in Section 3.4; capability benchmarks are summarized in the Discussion with full results in Appendix K.

The best classification layer is largely consistent across detectors. Table 3 reports the best classification layer and HC3 AI scores across all eight runs (standard CAA and three DW-CAA variants) $\times$ (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT). Across all detector-weighted runs 3-8, the detector used to extract the direction is used to score responses. Best classification layers are strikingly consistent across all three detectors, falling between layers 27-29 (75-81% of network depth) regardless of the weighting signal. This convergence suggests that all three detectors are responding to the same underlying representational structure in the residual stream despite their different characteristics.

Table 3: HC3 AI-probability scores. $(\pm\Delta)$ denotes change from baseline.

Run	Detector	Best Layer	Baseline	Best-layer Ablation	All-layer Ablation	Human Ref.
(1) CAA Base	RoBERTa	29	0.840	0.871 (+0.031)	0.651 (-0.189)	0.036
(2) CAA Instruct	RoBERTa	29	0.656	0.662 (+0.006)	0.196 (-0.460)	0.036
(3) RoBERTa Base	RoBERTa	29	0.843	0.833 (-0.010)	0.612 (-0.231)	0.036
(4) RoBERTa Instruct	RoBERTa	29	0.716	0.707 (-0.009)	0.132 (-0.584)	0.036
(5) GPTZero Base	GPTZero	28	0.687	0.480 (-0.207)	0.089 (-0.598)	0.005
(6) GPTZero Instruct	GPTZero	28	0.910	0.756 (-0.154)	0.093 (-0.817)	0.005
(7) RADAR Base	RADAR	29	0.989	0.963 (-0.026)	0.952 (-0.037)	0.677
(8) RADAR Instruct	RADAR	27	0.946	0.900 (-0.046)	0.872 (-0.074)	0.677

Ablation effectiveness across detectors. Table 3 reports RoBERTa HC3 AI-probability scores as a summary metric across all eight runs. LLM-judge AI-likeness scores are reported in full in Appendix E and show consistent trends. Under all-layer ablation, LLM-judge AI-likeness drops to the 2.15-2.90 range across all runs from baselines of 6.10-6.25, mirroring the pattern seen in the layer sweep and confirming that the direction extracted by each detector causally suppresses AI-style generation. Best-layer ablation produces near-zero change in LLM-judge AI-likeness across all six DW-CAA runs, which is consistent with the representation-causation dissociation from Section 3.2, confirming it is not an artifact of the standard CAA direction. Among detectors, GPTZero produces the most effective ablation. Both GPTZero runs show the largest HC3 score drops under all-layer ablation (base: -0.598; instruct: -0.817), suggesting that GPTZero's perplexity-based signals are particularly well-aligned with the generative causal subspace of the residual stream. RADAR is the least effective, with scores remaining very high across all conditions (-0.037 to -0.074 delta), likely reflecting a domain mismatch as RADAR was trained adversarially against Vicuna-7B paraphrases and may exploit features that do not correspond to those driving QWEN2.5-3B's stylistic generation.

3.4 The Style-Coherence Tradeoff: Pareto Analysis

A central practical question for any AI-style intervention is whether suppressing the style direction degrades output quality, and if so, where the optimal balance sits. We use writing coherence as the primary metric of interest, evaluated via LLM-as-judge scores on 20 MT-Bench-style writing prompts, because it is the most directly relevant capability for style interventions. To formally characterize and compare the style-coherency tradeoff across all intervention strategies, we apply the Pareto frontier analysis defined in Section 2: each condition is a point with $x = \mathrm{AI}_0 - \mathrm{AI}_i$ (suppression) and $y = C_i/C_0$ (coherence retention), and a condition is Pareto-optimal if no other achieves both greater suppression and higher coherence retention simultaneously. Full Pareto calculations are provided in Appendix J.

A tradeoff almost always exists. Across almost all ablation conditions, models, and detectors, suppressing the AI-style direction consistently reduces writing coherence to some degree. The magnitude varies substantially by intervention type, model, and writing category. All-layer ablation achieves the deepest suppression (reaching 2.05-2.70 on the 7-point scale) but at a coherence cost that makes many outputs practically unusable. Qualitative inspection reveals failure modes including repetitive sentence structures (e.g., *"I'm not a baby. I'm not a child. I'm not a teenager..."*) and structural fragmentation into numbered multiple-choice or bulleted list formats instead of prose. More examples can be found in Appendix G. Coherence costs are not uniform across writing categories: creative and narrative prompts preserve coherence better post-ablation, while professional and functional writing (resumes, cover letters, formal emails) degrades more heavily, consistent with functional formats being more tightly coupled to the AI-style direction. A full per-run category breakdown is provided in Appendix D.

The optimal tradeoff window sits in middle layers 18-25. Figures 1 and 2 show the Pareto frontiers for the eight ablation conditions in Table 3 and the layer sweep respectively. The layer sweep frontier (Figure 2) identifies where within the network the style–coherence tradeoff is most favorable. For QWEN2.5-3B with $k = 5$, the Pareto-optimal window sits around layers 18–25, where AI-likeness drops from a baseline of 6.2 to approximately 3.4 while writing coherence remains meaningfully above the all-layer ablation floor (≈2.0 vs. 1.6). For QWEN2.5-3B-INSTRUCT, the optimal window centers around layer 18 at $k = 5$, reaching AI-likeness = 3.2 with coherence = 2.8. The $k = 5$ windows consistently sit above $k = 1$ windows on the frontier: for any given suppression level, the 5-layer window achieves higher coherence retention, confirming that the style direction is causally distributed across several consecutive layers rather than sharply localized.

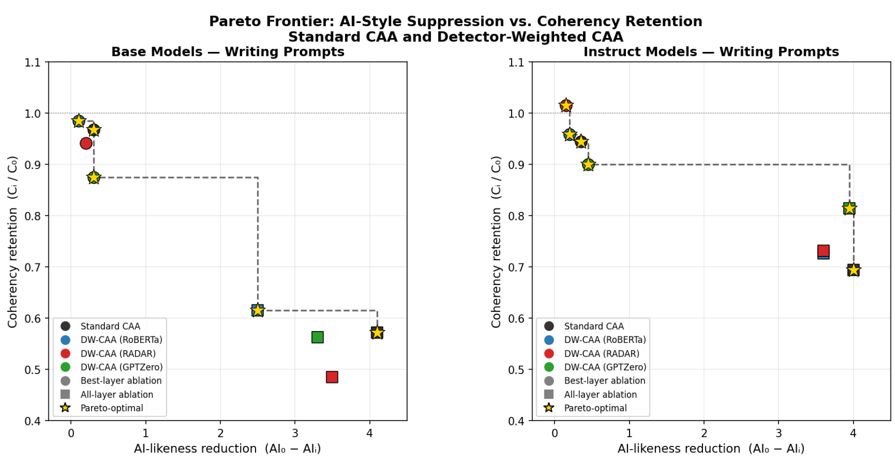

Figure 1: Pareto frontier for the eight ablation conditions in Table 3 on writing prompts, for QWEN2.5-3B (left) and QWEN2.5-3B-INSTRUCT (right). Each point is one ablation condition. Dashed line marks the Pareto frontier. The instruct frontier sits consistently higher than the base frontier, reflecting a more favorable tradeoff across all methods.

Suppression power is equivalent across all methods. In terms of AI-likeness suppression alone, detector-weighted CAA does not improve over standard CAA: all-layer ablation brings AI-likeness to the 2.10-2.15 range across all runs and both models regardless of direction extraction method. Best-layer ablation clusters near ($x \approx 0$, $y \approx 1$) across all runs, confirming that the representation-causation dissociation from Section 3.2 holds regardless of direction extraction method.

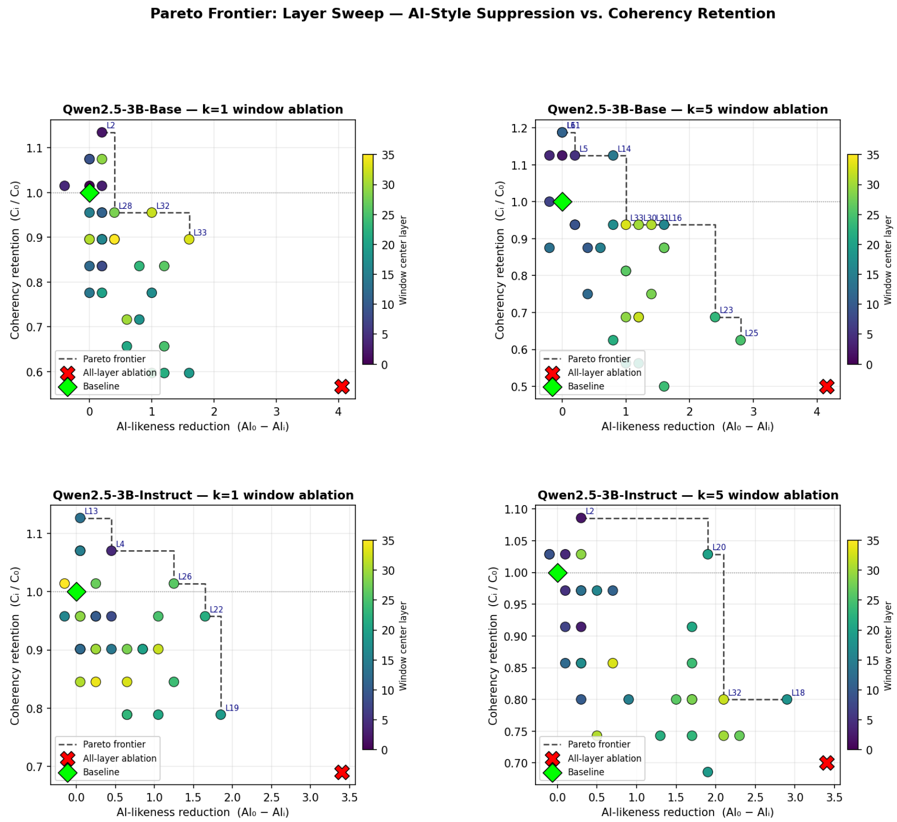

Figure 2: Pareto frontier for the layer sweep across all 36 window positions, for $k = 1$ and $k = 5$, on writing prompts. $k = 5$ windows consistently Pareto-dominate $k = 1$ windows. The instruct model frontier sits above the base model frontier at every suppression level.

DW-GPTZero achieves better coherence preservation in QWEN2.5-3B-INSTRUCT. The Pareto analysis reveals a distinction that suppression-only metrics miss. For QWEN2.5-3B-INSTRUCT, DW-GPTZero all-layer seems to be the dominant all-layer point on the frontier. It achieves suppression of 3.95 points while retaining 81.4% of baseline coherence, compared to standard CAA's 4.00 points of suppression at 69.4% coherence retention, which is a $\sim$17% improvement in coherence retention at almost equivalent suppression. One potential explanation is that GPTZero's perplexity-based decision boundary may be more precisely aligned with the stylistic component that is separable from coherence-preserving pathways in the RLHF-tuned residual stream. However, this advantage does not appear in QWEN2.5-3B, suggesting it is specific to the more factorized geometry that RLHF produces. This factorized geometry also explains the consistent advantage of QWEN2.5-3B-INSTRUCT over QWEN2.5-3B observed in Figures 1 and 2.

4 Discussion

AI-style versus refusal: a structural contrast. The representation-causation dissociation identified here stands in direct contrast with prior work on refusal [Arditi et al., 2024], where a single direction at one location mediates the behavior. For AI-style, the most linearly readable layer is causally inert as a window of middle layers is required for meaningful intervention. We hypothesize this difference reflects the distinct origins of the two properties. Refusal is a behaviorally trained binary switch, explicitly optimized by RLHF at a specific point in the network. AI-style, by contrast, is an emergent composite of many correlated generation patterns (formality, hedging, structure, and comprehensiveness) built up gradually across layers during pretraining on next-token prediction. This distributed origin makes it harder to localize and harder to ablate cleanly, and may explain why it does not admit the same single-direction, single-layer causal structure that refusal does.

Vocabulary signatures via logit lens. Projecting the AI-style direction onto the model's unembedding matrix reveals interpretable vocabulary clusters (see Appendix L for full details). Three clusters emerge on the AI side: emojis and markdown formatting tokens, formal Japanese constructions reflecting Qwen's multilingual pretraining, and web/code artifacts. The human-style tokens form a coherent contrasting cluster of informal English consisting of profanity, hedging qualifiers (`maybe`, `probably`, `apparently`), and emphatic colloquialisms, consistent with the HC3 human responses originating predominantly from the ELI5 subreddit.

Implications for AI text detection. The finding that AI style is causally generated in middle layers (15-22), rather than at the final readout layer, has practical implications for detector design. Detectors that probe final-layer representations may be reading a clean linear summary of a signal that was generated and committed several layers earlier. This raises the possibility that detection at intermediate layers could be both more causally informative and more robust to surface-level style transfer, since it would target the generative source rather than its downstream reflection.

Capability preservation. Ablating the AI-style direction is largely disjoint from general capabilities (full benchmark tables in Appendix K). MMLU and HellaSwag remain within ± 1.0 and ± 2.5 pp of baseline across all eight conditions, consistent with factual knowledge and commonsense reasoning being orthogonal to the style direction. TruthfulQA shows the largest effects under all-layer ablation—QWEN2.5-3B drops 8.0–9.0 pp while QWEN2.5-3B-INSTRUCT drops only 0.5–5.5 pp—a divergence consistent with RLHF encoding truthfulness more orthogonally to the style subspace. WikiText-2 perplexity follows a complementary split: it increases under ablation for QWEN2.5-3B but decreases under all-layer ablation for QWEN2.5-3B-INSTRUCT, reflecting that RLHF pushes the instruct model toward formal outputs that diverge from casual WikiText-2 prose, so removing the style direction moves it back toward the natural language distribution.

Limitations and future work. Several limitations constrain the scope of our conclusions. All experiments use a single model family (Qwen2.5-3B and its instruct variant), and cross-model generalization has not been tested. The causal layer profile and Pareto optimal style-coherency tradeoff window may differ across architectures and scales. The HC3 dataset contains only ChatGPT-generated AI text, so the extracted direction may not generalize to outputs from other LLMs such as Claude or Gemini, which have different stylistic signatures. LLM-as-judge evaluations are conducted with a maximum of 25 samples per window, resulting in high variance. Similarly, the Pareto coherence advantage of DW-GPTZero in QWEN2.5-3B-INSTRUCT should be interpreted with caution given the sample sizes and score variance involved. Replication with larger n per condition would be needed to confirm this as a robust effect. Conclusions about specific Pareto optimal layer positions should be treated as approximate rather than precise. Finally, the logit-lens vocabulary projection is correlational rather than causal. Future work should address these limitations through cross-model experiments to test generalization, sparse autoencoder decomposition of the style direction into interpretable features, and larger-sample evaluations to confirm the suggestive coherence advantage of DW-GPTZero in QWEN2.5-3B-INSTRUCT.

5 Conclusion

We have shown that the "AI-sounding" property of LLM outputs is genuinely linearly encoded in the residual stream independently of text length, but that this encoding is causally distributed in a way that creates a sharp dissociation between representation and generation. Suppressing the direction consistently reduces AI-likeness but always at some coherence cost, and a Pareto frontier analysis identifies the optimal intervention window (layers 18–25, $k = 5$) and reveals that DW-GPTZero achieves $\sim$17% better coherence retention than standard CAA in QWEN2.5-3B-INSTRUCT. Factual and commonsense capabilities are largely preserved under ablation, while TruthfulQA and perplexity reveal a systematic QWEN2.5-3B-QWEN2.5-3B-INSTRUCT divergence. These findings speak to a broader challenge in mechanistic interpretability: the tools we use to understand model internals (linear probing, difference-in-means directions, classification accuracy by layer) measure where information is *readable*, not where it *acts*. As LLMs become embedded in consequential domains from academic integrity to content moderation, interventions designed to suppress or detect AI-style behavior must target the generative source rather than its downstream reflection. More broadly, understanding how stylistic, behavioral, and factual properties are encoded and causally distributed across transformer layers is a prerequisite for developing reliable, principled methods for model steering, AI detection, and alignment.

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A Length Confound on Original HC3

Figures 3–5 illustrate the verbosity confound present in the original HC3 dataset. On the original data, the extracted CAA direction has cosine similarity 0.938 with a pure text-length direction, indicating that most of the classification signal is attributable to response length rather than style. After length-matching (Section 3.1), this confound is eliminated.

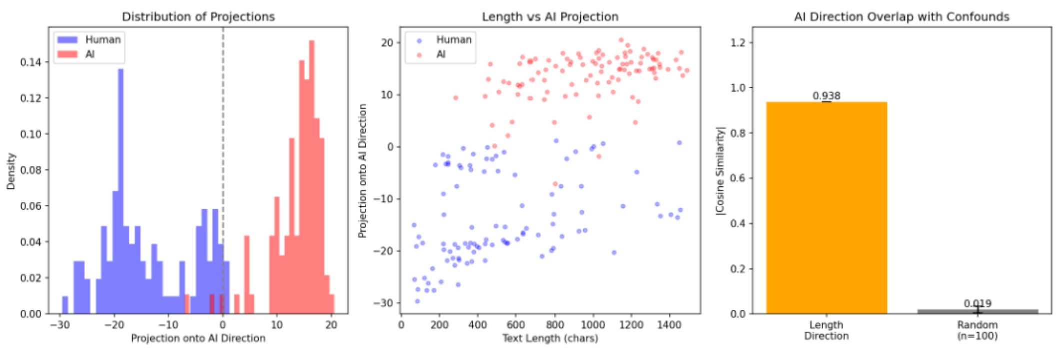

Figure 3: (1) QWEN2.5-3B (original HC3): (left) projection distributions onto the extracted CAA direction; (center) text length vs. projection score, revealing that the direction tracks response verbosity; (right) cosine similarity of the AI direction with the length direction (0.938) versus random directions.

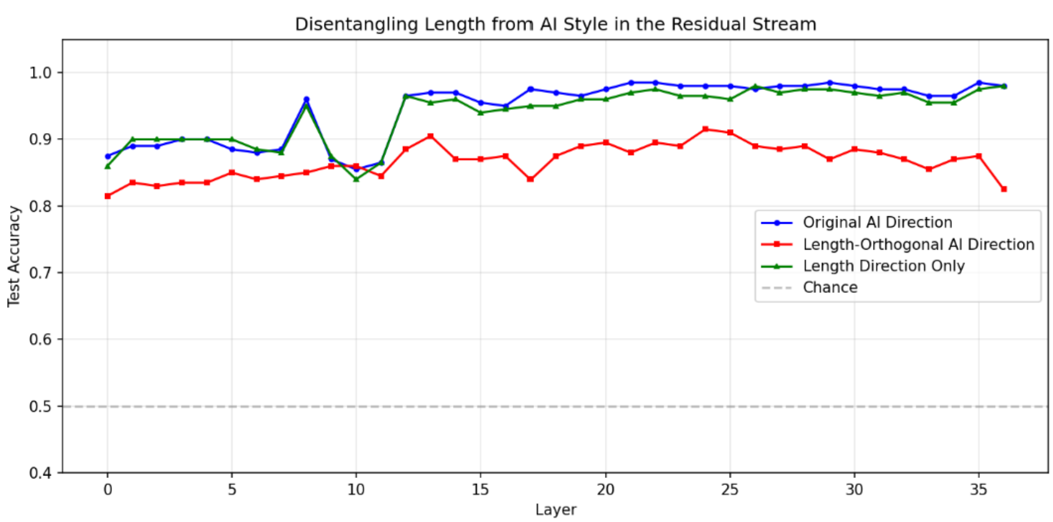

Figure 4: (1) QWEN2.5-3B (original HC3): classification accuracy by layer for the original AI direction, the length-orthogonal AI direction, and the length direction alone. The length direction alone achieves near-identical accuracy to the full AI direction, confirming verbosity as the dominant signal.

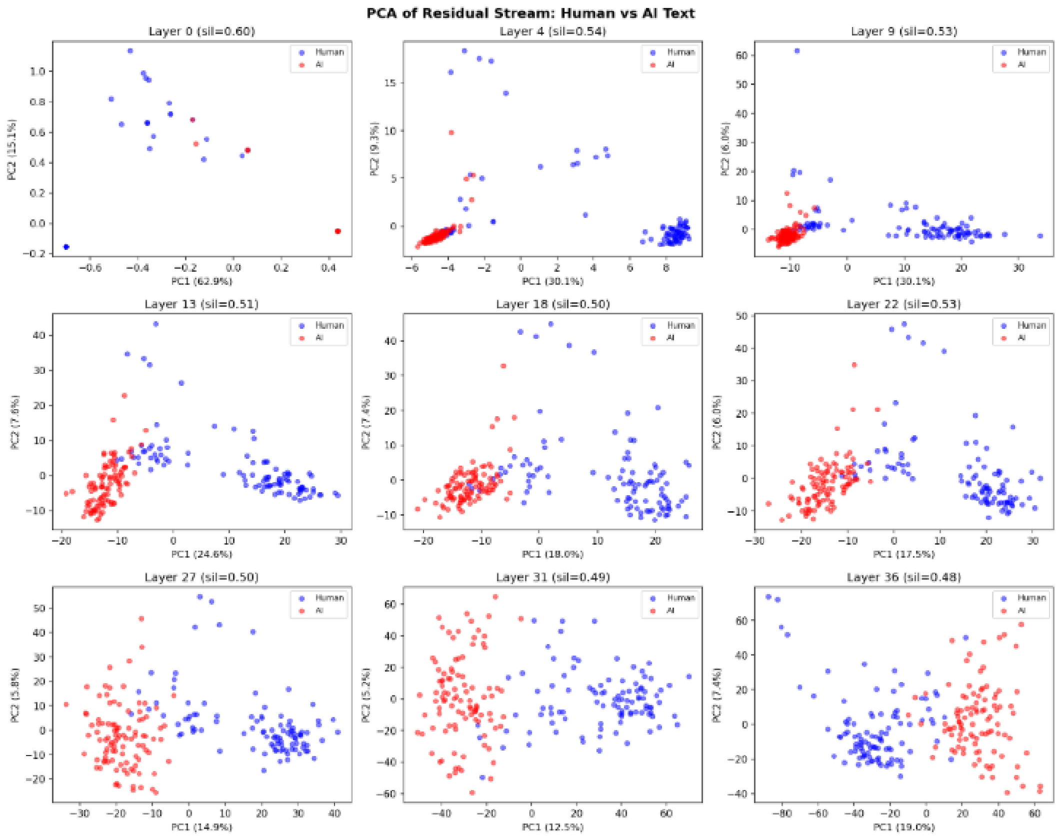

Figure 5: PCA of residual stream activations for (1) QWEN2.5-3B (original HC3) across selected layers. Human (blue) and AI (red) clusters separate clearly, but this separation is substantially driven by the length confound.

B Length-Matched HC3: Direction Visualizations

Figures 6–8 show the direction analyses on the length-matched HC3 dataset, corresponding to the results described in Section 3.1. These complement the original HC3 confound figures in Appendix A.

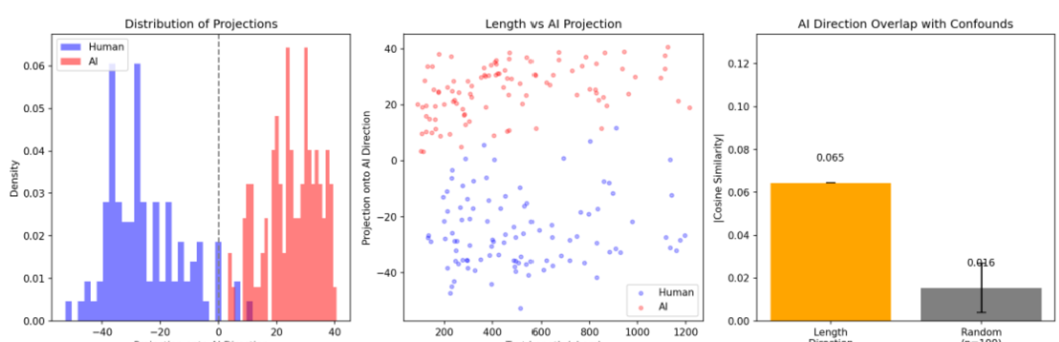

Figure 6: (3) QWEN2.5-3B (length-matched): (left) projection distributions onto the extracted CAA direction; (center) text length vs. projection score, showing no relationship between length and the extracted direction; (right) cosine similarity with the length direction collapses to -0.065, near that of random directions.

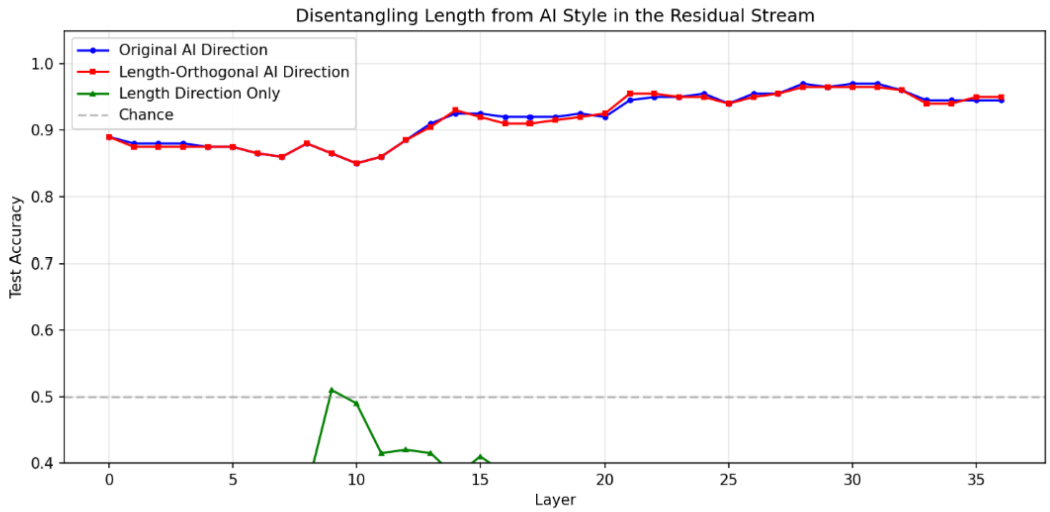

Figure 7: (3) QWEN2.5-3B (length-matched): classification accuracy by layer for the original AI direction and the length-orthogonal AI direction (which now nearly coincide), alongside the length-only direction, which collapses to below-chance accuracy, confirming length carries no discriminative signal after matching.

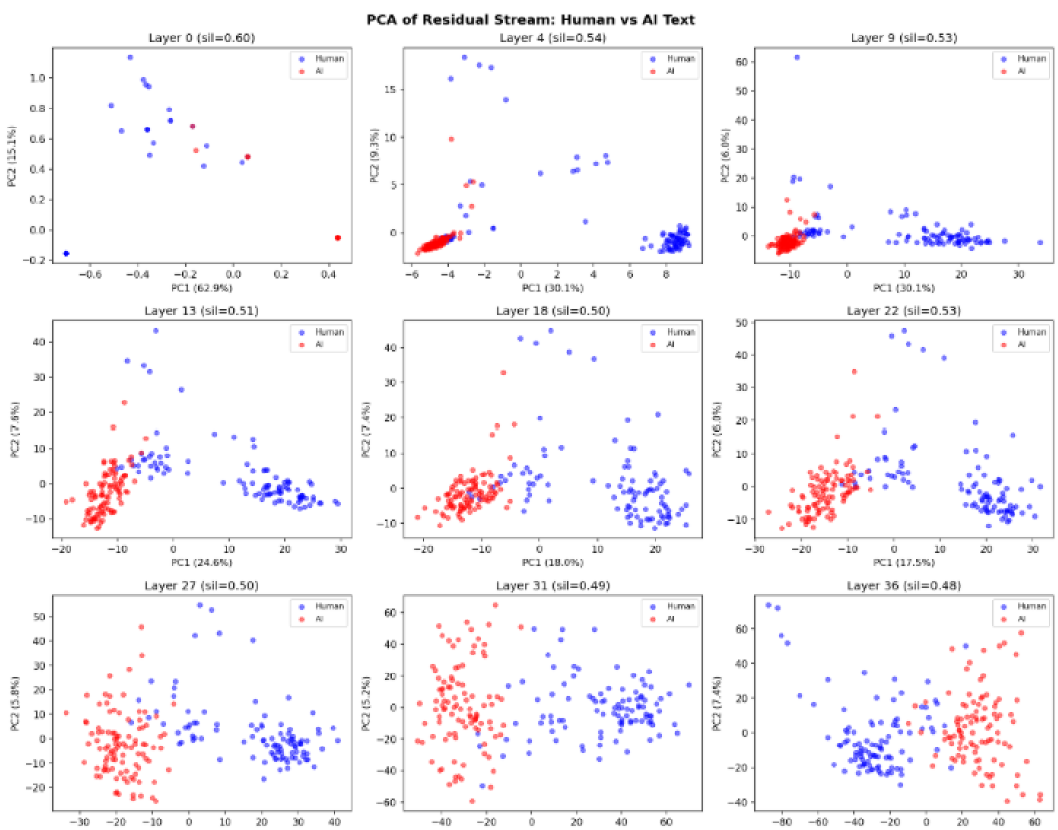

Figure 8: PCA of residual stream activations for (3) QWEN2.5-3B (length-matched) across selected layers. Human and AI clusters remain well separated, confirming that the linear structure reflects genuine stylistic differences rather than verbosity.

C Layer Sweep: Visualization

Figure 9 shows the full sliding-window sweep across all 36 layer positions for QWEN2.5-3B (blue) and QWEN2.5-3B-INSTRUCT (red), with $k = 1$ (left column) and $k = 5$ (right column). Each row shows a different evaluation metric: LLM-judge AI-likeness (top), writing quality (second), factual quality (third), and RoBERTa HC3 AI score (bottom). Dotted horizontal lines mark the unablated baseline; dash-dot lines mark all-layer ablation. Shaded regions are ± 1 SD. The best-classifying layer (28-29, gray dashed vertical line) consistently produces near-zero change across all metrics, while the causal generation region at middle layers produces the largest reductions in AI-likeness. Full per-window numerical results are in Appendix F.

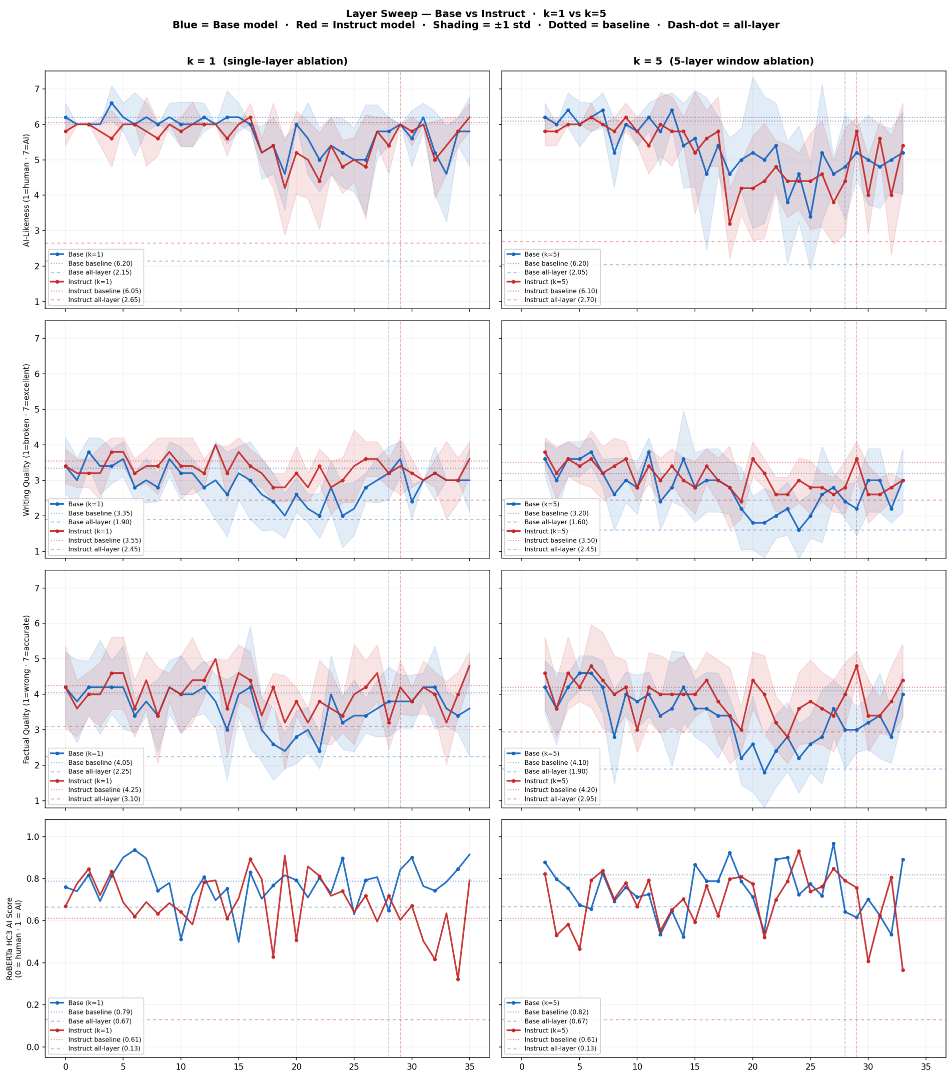

Figure 9: Layer sweep ablation results across all 36 layers for QWEN2.5-3B (blue) and QWEN2.5-3B-INSTRUCT (red), with $k = 1$ (left column) and $k = 5$ (right column). The best-classifying layer (28-29, gray dashed vertical line) consistently produces near-zero change, while the causal generation region at middle layers produces the largest reductions in AI-likeness.

D Writing Quality by Category

Figures 10 and 11 show the per-category breakdown of ablation effects across all six detector-weighted runs. Figure 10 plots the drop in AI-likeness and coherence (baseline minus all-layer ablation) for each writing category, making the differential costs visible across detectors. Figure 11 shows absolute AI-likeness and coherence scores under all-layer ablation as a heatmap, with rows corresponding to runs and columns to writing categories.

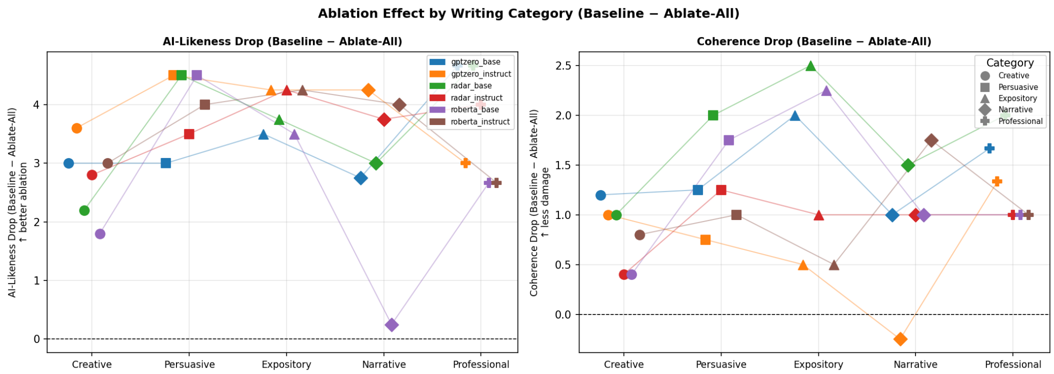

Figure 10: AI-likeness drop (left) and coherence drop (right) from baseline to all-layer ablation, broken down by writing category and detector run. Higher values indicate larger reductions. Creative and narrative categories show smaller coherence drops while persuasive and expository categories show larger AI-likeness reductions. Professional/functional writing is the most resistant to AI-likeness reduction, consistent with its structural formatting being tightly coupled to the style direction.

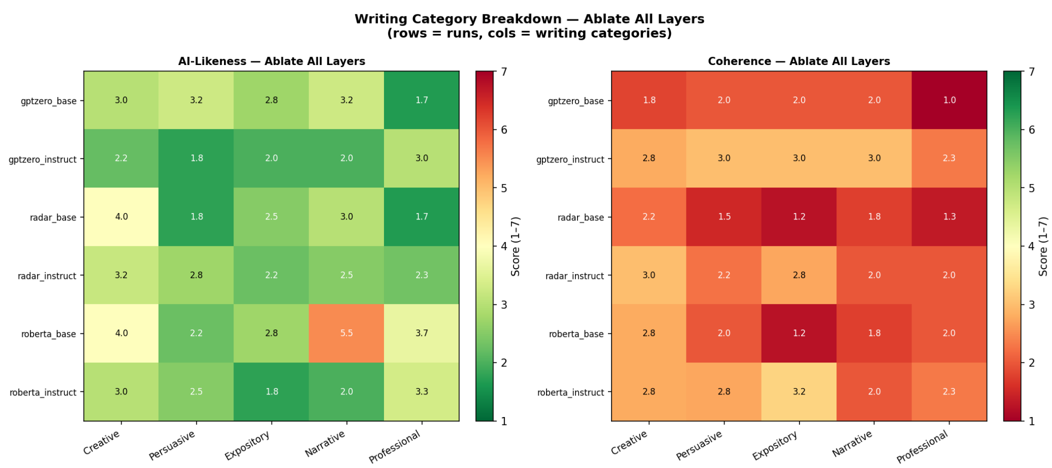

Figure 11: Writing category breakdown under all-layer ablation across all six detector-weighted runs (rows) and five writing categories (columns). Left: AI-likeness scores (green = low = more human-like); right: coherence scores (green = high = better preserved). Professional/functional writing retains the highest AI-likeness scores post-ablation, while expository and persuasive categories achieve the deepest AI-likeness reductions. Coherence is most preserved in narrative and creative categories.

E Standard CAA and Detector-Weighted CAA: Full LLM-Judge Results

Tables 4–7 report full LLM-judge scores (AI-likeness, coherence, factual) for standard CAA and all six detector-weighted runs across baseline, best-layer ablation, and all-layer ablation conditions, for both HC3 and writing prompt evaluations.

Table 4: LLM-judge scores on HC3 prompts — base models ($n = 25$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
Standard CAA Base	Baseline	5.48±0.98	2.44±1.30	2.44±1.44
	Best-layer ablation	4.20±1.44	2.44±1.20	2.20±1.10
	All-layer ablation	2.08±1.29	1.24±0.59	1.24±0.51
RoBERTa Base	Baseline	5.52±1.06	3.16±1.59	3.00±1.67
	Best-layer ablation	4.56±1.30	2.44±1.33	2.08±1.29
	All-layer ablation	2.64±1.60	1.88±0.99	2.04±1.04
RADAR Base	Baseline	5.24±1.14	3.20±1.30	2.92±1.44
	Best-layer ablation	4.68±1.32	2.40±1.33	2.20±1.20
	All-layer ablation	2.60±1.67	1.32±0.55	1.60±1.17
GPTZero Base	Baseline	5.24±1.34	2.88±1.51	2.84±1.62
	Best-layer ablation	3.96±1.68	2.20±0.80	2.32±1.22
	All-layer ablation	2.28±1.31	1.32±0.61	1.36±0.74

Table 5: LLM-judge scores on HC3 prompts — instruct models ($n = 25$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
Standard CAA Instruct	Baseline	5.60±0.80	2.84±1.01	2.88±1.27
	Best-layer ablation	5.00±1.36	2.84±0.73	2.56±0.70
	All-layer ablation	2.64±0.93	2.08±0.39	2.20±0.85
RoBERTa Instruct	Baseline	5.48±0.98	3.36±0.97	3.00±1.13
	Best-layer ablation	4.76±1.14	2.88±1.11	2.64±1.09
	All-layer ablation	2.48±1.20	2.24±0.86	2.12±1.21
RADAR Instruct	Baseline	5.80±0.89	3.04±0.82	2.64±1.09
	Best-layer ablation	5.28±1.00	2.80±0.75	2.68±1.26
	All-layer ablation	2.56±0.85	2.60±0.75	2.36±0.84
GPTZero Instruct	Baseline	5.68±0.93	3.48±1.06	3.68±1.52
	Best-layer ablation	4.72±1.25	2.72±0.87	2.80±0.98
	All-layer ablation	2.16±0.73	2.08±0.56	2.20±0.85

Table 6: LLM-judge scores on writing prompts — base models ($n = 20$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
Standard CAA Base	Baseline	6.20±0.40	3.15±0.65	4.05±0.97
	Best-layer ablation	5.90±0.44	3.05±0.67	3.70±0.84
	All-layer ablation	2.10±1.58	1.80±0.81	2.15±1.01
RoBERTa Base	Baseline	6.15±0.57	3.25±0.83	4.10±0.99
	Best-layer ablation	6.05±0.59	3.20±0.68	3.80±0.98
	All-layer ablation	3.65±1.74	2.00±0.95	2.60±1.16
RADAR Base	Baseline	6.20±0.40	3.40±0.58	4.10±0.77
	Best-layer ablation	6.00±0.71	3.20±0.75	3.75±1.04
	All-layer ablation	2.70±1.73	1.65±0.73	2.00±0.77
GPTZero Base	Baseline	6.15±0.48	3.20±0.60	4.10±0.77
	Best-layer ablation	5.85±0.65	2.80±0.87	3.55±1.16
	All-layer ablation	2.85±1.74	1.80±0.87	2.10±0.99

Table 7: LLM-judge scores on writing prompts — instruct models ($n = 20$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
Standard CAA Instruct	Baseline	6.15±0.36	3.60±0.58	4.50±0.87
	Best-layer ablation	5.80±0.40	3.40±0.66	4.00±0.95
	All-layer ablation	2.15±1.11	2.50±0.59	3.20±0.98
RoBERTa Instruct	Baseline	6.10±0.44	3.65±0.48	4.45±0.80
	Best-layer ablation	5.90±0.54	3.50±0.81	4.15±0.85
	All-layer ablation	2.50±1.24	2.65±0.65	3.30±0.90
RADAR Instruct	Baseline	6.25±0.43	3.35±0.57	4.25±0.83
	Best-layer ablation	6.10±0.54	3.40±0.58	4.10±0.70
	All-layer ablation	2.65±1.24	2.45±0.80	2.95±1.02
GPTZero Instruct	Baseline	6.10±0.44	3.50±0.81	4.40±0.86
	Best-layer ablation	5.65±0.73	3.15±0.65	3.90±0.99
	All-layer ablation	2.15±0.85	2.85±0.36	3.35±0.73

F Layer Sweep: Full Window Results

Tables 8–11 report LLM-judge AI-likeness, coherence, and factual scores for every window position in the layer sweep, for all four conditions (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT) × ($k = 1$, $k = 5$). Reference conditions (baseline and all-layer ablation) are included at the top of each table. $n = 5$ per window position.

Table 8: Layer sweep results: QWEN2.5-3B, $k = 5$. Best classification layer: 28. Baseline: AI=6.20, Coh=3.20, Fact=4.10. All-layer: AI=2.05, Coh=1.60, Fact=1.90.

Window	Center	AI-likeness	Coherence	Factual
0	2	6.20±0.40	3.60±0.49	4.20±0.75
1	3	6.00±0.00	3.00±0.89	3.60±1.02
2	4	6.40±0.49	3.60±0.49	4.20±0.40
3	5	6.00±0.63	3.60±0.49	4.60±0.49
4	6	6.20±0.40	3.80±0.40	4.60±0.49
5	7	6.40±0.49	3.20±0.40	4.20±0.75
6	8	5.20±0.98	2.60±1.02	2.80±1.33
7	9	6.00±0.00	3.00±0.63	4.00±0.63
8	10	5.80±0.40	2.80±0.75	3.80±0.75
9	11	6.20±0.40	3.80±0.40	4.00±0.63
10	12	5.80±0.98	2.40±0.80	3.40±0.49
11	13	6.40±0.49	2.80±0.40	3.60±1.62
12	14	5.40±1.20	3.60±1.36	4.20±0.98
13	15	5.60±1.36	2.80±0.98	3.60±0.80
14	16	4.60±2.15	3.00±0.89	3.60±1.02
15	17	5.40±0.80	3.00±0.89	3.40±1.20
16	18	4.60±1.02	2.80±0.75	3.40±0.49
17	19	5.00±0.89	2.20±1.17	2.20±0.75
18	20	5.20±2.13	1.80±0.75	2.60±1.36
19	21	5.00±1.79	1.80±0.98	1.80±0.98
20	22	5.40±1.20	2.00±0.63	2.40±1.02
21	23	3.80±1.72	2.20±0.75	2.80±0.98
22	24	4.60±1.36	1.60±0.80	2.20±0.98
23	25	3.40±1.50	2.00±0.63	2.60±0.80
24	26	5.20±1.94	2.60±1.36	2.80±1.33
25	27	4.60±0.49	2.80±0.40	3.60±0.80
26	28	4.80±1.47	2.40±0.49	3.00±1.09
27	29	5.20±0.75	2.20±0.75	3.00±0.63
28	30	5.00±1.26	3.00±0.89	3.20±0.75
29	31	4.80±1.17	3.00±0.89	3.40±1.20
30	32	5.00±0.89	2.20±0.40	2.80±0.75
31	33	5.20±1.17	3.00±0.89	4.00±0.63

Table 9: Layer sweep results: QWEN2.5-3B-INSTRUCT, $k = 5$. Best classification layer: 29. Baseline: AI=6.10, Coh=3.50, Fact=4.20. All-layer: AI=2.70, Coh=2.45, Fact=2.95.

Window	Center	AI-likeness	Coherence	Factual
0	2	5.80±0.40	3.80±0.40	4.60±1.02
1	3	5.80±0.40	3.20±0.75	3.60±0.80
2	4	6.00±0.00	3.60±0.49	4.60±1.02
3	5	6.00±0.00	3.40±0.49	4.20±0.40
4	6	6.20±0.40	3.60±0.80	4.80±1.17
5	7	6.00±0.00	3.20±0.75	4.40±1.36
6	8	5.80±0.40	3.40±0.80	4.00±1.09
7	9	6.20±0.40	3.60±0.49	4.20±0.75
8	10	5.80±0.40	2.80±0.40	3.00±0.63
9	11	5.40±0.80	3.40±0.49	4.20±0.98
10	12	6.00±0.89	3.00±0.63	4.00±1.09
11	13	5.80±0.98	3.40±0.49	4.00±0.89
12	14	5.80±0.40	3.00±0.63	4.00±1.09
13	15	5.20±1.72	2.80±0.75	4.00±0.63
14	16	5.60±0.80	3.40±0.49	4.40±0.80
15	17	5.80±0.98	3.00±0.63	3.80±1.17
16	18	3.20±0.98	2.80±1.17	3.40±1.36
17	19	4.20±0.75	2.40±0.49	3.00±0.63
18	20	4.20±1.47	3.60±0.49	4.40±0.80
19	21	4.40±1.62	3.20±0.40	4.00±1.09
20	22	4.80±0.75	2.60±0.49	3.20±0.75
21	23	4.40±1.02	2.60±0.49	2.80±0.75
22	24	4.40±0.80	3.00±0.63	3.60±1.02
23	25	4.40±1.36	2.80±0.75	3.80±1.17
24	26	4.60±1.50	2.80±0.40	3.60±1.02
25	27	3.80±1.17	2.60±0.49	3.40±1.02
26	28	4.40±1.50	2.80±0.75	4.00±0.89
27	29	5.80±0.40	3.60±0.49	4.80±0.40
28	30	4.00±1.09	2.60±0.80	3.40±1.02
29	31	5.60±0.49	2.60±0.49	3.40±0.49
30	32	4.00±1.67	2.80±0.40	3.80±0.98
31	33	5.40±1.20	3.00±0.00	4.40±1.02

Table 10: Layer sweep results: QWEN2.5-3B, $k = 1$. Best classification layer: 28. Baseline: AI=6.20, Coh=3.35, Fact=4.05. All-layer: AI=2.15, Coh=1.90, Fact=2.25.

Window	Center	AI-likeness	Coherence	Factual
0	0	6.20±0.40	3.40±0.80	4.20±0.98
1	1	6.00±0.00	3.00±0.63	3.80±1.17
2	2	6.00±0.00	3.80±0.40	4.20±0.75
3	3	6.00±0.00	3.40±0.80	4.20±1.33
4	4	6.60±0.49	3.40±0.49	4.20±0.75
5	5	6.20±0.40	3.60±0.49	4.20±1.17
6	6	6.00±0.89	2.80±0.40	3.40±0.49
7	7	6.20±0.40	3.00±0.63	3.80±0.40
8	8	6.00±0.00	2.80±0.40	3.40±1.02
9	9	6.20±0.40	3.60±0.49	4.20±0.75
10	10	6.00±0.63	3.20±0.75	4.00±1.09
11	11	6.00±0.63	3.20±0.40	4.00±0.63
12	12	6.20±0.40	2.80±0.40	4.20±0.75
13	13	6.00±0.00	3.00±1.09	3.80±0.75
14	14	6.20±0.75	2.60±1.20	3.00±1.41
15	15	6.20±0.40	3.20±0.75	4.00±0.63
16	16	6.00±0.00	3.00±1.09	4.20±1.72
17	17	5.20±0.75	2.60±1.02	3.00±0.89
18	18	5.40±0.80	2.40±0.80	2.60±1.02
19	19	4.60±1.02	2.00±0.63	2.40±0.49
20	20	6.00±0.00	2.60±0.49	2.80±0.75
21	21	5.60±1.02	2.20±0.40	3.00±0.63
22	22	5.00±0.89	2.00±0.63	2.40±0.49
23	23	5.40±0.80	2.80±0.75	4.00±1.09
24	24	5.20±0.98	2.00±0.89	3.20±0.75
25	25	5.00±0.89	2.20±0.75	3.40±1.02
26	26	5.00±1.55	2.80±0.40	3.40±0.49
27	27	5.80±0.75	3.00±0.63	3.60±0.80
28	28	5.80±0.40	3.20±0.75	3.80±0.98
29	29	6.00±0.00	3.60±0.49	3.80±0.75
30	30	5.60±0.80	2.40±0.49	3.80±0.75
31	31	6.20±0.40	3.00±0.00	4.20±0.40
32	32	5.20±1.17	3.20±0.75	4.20±1.17
33	33	4.60±1.36	3.00±0.00	3.60±0.49
34	34	5.80±0.40	3.00±0.00	3.40±0.49
35	35	5.80±0.98	3.00±0.89	3.60±1.36

Table 11: Layer sweep results: QWEN2.5-3B-INSTRUCT, $k = 1$. Best classification layer: 29. Baseline: AI=6.05, Coh=3.55, Fact=4.25. All-layer: AI=2.65, Coh=2.45, Fact=3.10.

Window	Center	AI-likeness	Coherence	Factual
0	0	5.80±0.40	3.40±0.49	4.20±1.17
1	1	6.00±0.00	3.20±0.40	3.60±0.80
2	2	6.00±0.00	3.20±0.40	4.00±0.63
3	3	5.80±0.40	3.20±0.75	4.00±0.89
4	4	5.60±0.80	3.80±0.40	4.60±1.02
5	5	6.00±0.00	3.80±0.40	4.60±1.02
6	6	6.00±0.00	3.20±0.40	3.60±0.80
7	7	5.80±0.98	3.40±0.49	4.40±0.80
8	8	5.60±0.49	3.40±0.80	3.40±1.36
9	9	6.00±0.00	3.80±0.40	4.20±0.40
10	10	5.80±0.40	3.40±0.80	4.00±1.09
11	11	6.00±0.63	3.40±0.80	4.40±1.20
12	12	6.00±0.00	3.20±0.40	4.40±0.49
13	13	6.00±0.00	4.00±0.00	5.00±0.00
14	14	5.60±0.49	3.20±0.75	3.60±1.36
15	15	6.00±0.00	3.80±0.40	4.60±0.80
16	16	6.20±0.40	3.40±0.49	4.40±0.80
17	17	5.20±0.40	3.20±0.40	3.40±0.49
18	18	5.40±1.20	2.80±0.40	4.20±0.40
19	19	4.20±1.33	2.80±0.40	3.20±1.33
20	20	5.20±1.17	3.20±0.75	3.80±0.40
21	21	5.00±1.09	2.80±0.40	3.20±0.40
22	22	4.40±1.36	3.40±0.49	3.80±1.17
23	23	5.40±0.80	2.80±0.75	3.60±1.02
24	24	4.80±0.75	3.00±0.63	3.40±0.49
25	25	5.00±0.63	3.40±1.02	4.00±1.41
26	26	4.80±1.47	3.60±0.49	4.20±0.75
27	27	5.80±0.40	3.60±0.49	4.60±0.80
28	28	5.40±0.80	3.20±0.40	3.20±0.75
29	29	6.00±0.00	3.40±0.80	4.20±0.75
30	30	5.80±0.40	3.20±0.40	3.80±0.40
31	31	6.00±0.00	3.00±0.00	4.20±0.75
32	32	5.00±1.09	3.20±0.40	4.00±0.63
33	33	5.40±0.80	3.00±1.09	3.20±1.17
34	34	5.80±0.40	3.00±0.63	4.00±0.63
35	35	6.20±0.40	3.60±0.49	4.80±0.40

G Ablated Writing Samples

The following examples illustrate qualitative output changes under all-layer ablation for the GPTZero QWEN2.5-3B-INSTRUCT run, selected to represent three characteristic patterns: coherence collapse in creative writing, degradation in explanatory writing, and resistance to ablation in professional formats.

Creative writing: poem about rain. *Prompt: "Write a short poem about the feeling of rain on a summer evening."*

Baseline (GPTZero score: 1.000):

> *The gentle rain starts to fall, / On a summer evening's calm. / It brushes against my skin, / A cool, refreshing feel. / The air grows fresh and clean, / As droplets dance in light. / I close my eyes and breathe it in, / This moment, so divine. / The world around me seems to slow, / As nature's symphony plays. / And I am lost in its embrace, / In this quiet, peaceful haze.*

All-layer ablation (GPTZero score: 0.003):

> *Rain on the summery eve / Drops of purest, softest glee / Lullaby of a summer night / I'll not be going out tonight / The cool air in my nostrils fills / Bursts of life through every cell / A world so very nice / [. . .] Oh, Summer! I can't wait till next time / To feel your soft, wet kiss on me / A new day just begins / And you are here to stay* [output trails off mid-thought]

The ablated output shifts from structured, balanced stanzas toward fragmented, more colloquial phrasing. The GPTZero score drops from 1.000 to 0.003, reflecting a near-complete ablation of the perplexity-based AI signal, but the output also loses the formal prosodic structure of the baseline.

Explanatory writing: concept of time. *Prompt: "Explain the concept of time to someone who has never encountered the idea."*

Baseline (GPTZero score: 1.000):

> *Time is a measure of duration or progression, often measured in units such as seconds, minutes, hours, days, weeks, months, and years. It is a fundamental concept that helps us organize our lives and understand the passage of events. [. . .] Time can also be thought of as a flow, like water flowing through a stream. Just as the water moves from one place to another, so does time move from one moment to the next.*

All-layer ablation (GPTZero score: 0.002):

> *Time is a thing that can be measured and it goes forward, and can not be reversed. [. . .] I am 31 years old means I have been alive for 31 years [. . .] You are always 10 minutes younger than you were last week. You are a 12 year old boy. Imagine you have a friend that has a bike, you ask him if he could lend you his bike and ride with you to get ice cream, you will say something like "when are you free?"*

The ablated output becomes informal and concrete but retains topical relevance. The shift toward colloquial examples (bike rides, ice cream) is consistent with the model reverting to Reddit/ELI5-style text under ablation.

Professional writing: job offer decline email. *Prompt: "Write a professional email declining a job offer while maintaining a positive relationship."*

Baseline (GPTZero score: 0.999):

> *Subject: Declining the Job Offer / Dear [Hiring Manager's Name], / I hope this message finds you well. I am writing to express my gratitude for the opportunity to interview and consider a position with [Company Name]. [. . .] After careful consideration of all aspects, including the details of the offer, my current situation, and some recent personal developments, I have decided that it would be best for me to decline the position at this time.*

All-layer ablation (GPTZero score: 0.997):

> *Subject: No thanks for the position / Hi [Hr name] / I have been thinking about your job offer for quite some time, and I really appreciate the opportunity. After much deliberation, I have decided to not take up this offer. [...] So please accept my apologies for not accepting this offer but I hope you understand and that we can stay as friends / colleagues etc.*

This example illustrates the professional/functional category's resistance to ablation: despite all-layer projection, the GPTZero score barely moves (0.999 $\rightarrow$ 0.997). The output does shift from formal to casual register ("Hi [Hr name]", "friends / colleagues etc."), but the email structure itself persists, consistent with the finding in Appendix D that professional formats are among the most tightly coupled to the AI-style direction.

H LLM-as-Judge Prompts

HC3 / ELI5 Evaluation Prompt

> You are evaluating responses to questions from the HC3 dataset (Reddit's r/explainlikeimfive subreddit). These are questions where people ask for accessible explanations of complex topics. Responses can range from casual Reddit-style answers to formal AI-generated explanations.
>
> Original question: {prompt}
> Generated response: {response}
>
> Rate this response on three dimensions using the anchored scales below.
>
> **AI-LIKENESS (1–7):**
> 1 = Clearly human: casual, informal, conversational, short, may have typos
> 2 = Mostly human: informal but structured
> 3 = Slightly human: some informal elements, some structure
> 4 = Mixed: formal and informal elements in roughly equal measure
> 5 = Slightly AI: mostly formal/structured with occasional casual phrasing
> 6 = Mostly AI: formal, comprehensive, well-organised, hedging language
> 7 = Clearly AI: highly formal, bullet-pointed, "I'd be happy to help", comprehensive coverage, textbook-style explanations
>
> **COHERENCE (1–7):**
> 1 = Completely incoherent, off-topic, or clearly broken (e.g. repeating text, format collapse into multiple-choice questions)
> 4 = Mostly on-topic but with noticeable gaps or tangents
> 7 = Clear, on-topic, well-organised, directly answers the question
>
> **FACTUAL REASONABLENESS (1–7):**
> 1 = Clearly wrong, nonsensical, or contains major factual errors
> 4 = Mostly correct with minor inaccuracies
> 7 = Accurate, informative, no obvious errors
>
> Respond with ONLY a JSON object, no other text: {"ai_likeness": <1-7>, "coherence": <1-7>, "factual": <1-7>}

Writing Prompt Evaluation

> You are evaluating responses to creative and expository writing prompts.
>
> Writing prompt: {prompt}
> Generated response: {response}
>
> **AI-LIKENESS (1–7):**
> 1 = Clearly human: casual, idiosyncratic, natural imperfections, personal voice
> 2 = Mostly human: personal voice with minor formal elements
> 3 = Slightly human: some natural phrasing but noticeable polish
> 4 = Mixed: some human touches, some AI-typical patterns in roughly equal measure
> 5 = Slightly AI: mostly polished/structured with occasional natural phrasing
> 6 = Mostly AI: uniformly polished, generic metaphors, hedging, over-completeness
> 7 = Clearly AI: highly formulaic structure, exhaustive coverage, "certainly" / "I'd be happy to", perfectly balanced sentences, textbook-style prose
>
> **WRITING QUALITY (1–7):**
> 1 = Completely broken: incoherent, off-topic, format collapse, repetition

2 = Very poor: mostly off-topic or barely readable
3 = Poor: several noticeable problems but some coherence
4 = Adequate: addresses the prompt but writing is flat or generic
5 = Good: addresses the prompt with reasonable style and clarity
6 = Very good: engaging, clear, shows craft in language choice
7 = Excellent: vivid, distinctive, well-crafted, effectively addresses the prompt

FACTUAL / CREATIVE REASONABLENESS (1–7):
For factual writing: 1 = clear errors, 7 = accurate and well-grounded
For creative writing: 1 = incoherent or contradictory, 7 = internally consistent, imaginative, and effective

Respond with ONLY a JSON object, no other text: {"ai_likeness": <1-7>, "coherence": <1-7>, "factual": <1-7>}

I Writing Evaluation Prompts

The following 20 MT-Bench-style prompts were used for all writing evaluations, spanning five categories of four prompts each.

Creative Writing.

1. Write a short poem about the feeling of rain on a summer evening.
2. Write the opening paragraph of a mystery novel set in a lighthouse.
3. Write a short story about a child who discovers that their shadow has a mind of its own.
4. Create a myth explaining why the sky changes colors at sunset.
5. Write a product review for the concept of ‘sleep’ as if it were a new consumer technology.

Persuasive / Argumentative.

6. Write a persuasive argument for why everyone should learn to cook their own meals.
7. Write a speech encouraging high school students to pursue careers in science.
8. Write a brief manifesto for a society that values rest and slowness over productivity.
9. Write a letter to your city council arguing that more public green spaces should be built.

Expository / Explanatory.

10. Explain the concept of time to someone who has never encountered the idea.
11. Describe what makes a city feel alive, drawing on sensory and social details.
12. Explain how music can change a person’s emotional state, using concrete examples.
13. Describe the water cycle as if you were telling a story to a curious child.

Narrative / Reflective.

14. Write a letter from a tree to the person who planted it fifty years ago.
15. Write a short scene where two strangers find common ground while stuck in an elevator.
16. Write a reflection on what you would tell your future self in a time capsule.
17. Describe a moment of unexpected kindness you witnessed or imagined.

Professional / Functional.

18. Write a professional email declining a job offer while maintaining a positive relationship.
19. Write a recipe for ‘making new friends’ formatted as a cooking recipe with ingredients and steps.
20. Write a product launch announcement for a new renewable energy startup called ‘Solara’.

J Pareto Frontier: Full Calculations

The Pareto frontier analysis places each ablation condition on a common two-dimensional axis: $x = \mathrm{AI}_0 - \mathrm{AI}_i$ (AI-likeness suppression, higher is better) and $y = C_i/C_0$ (coherence retention, higher is better), where AI_0 and C_0 are the unablated baseline scores. A condition is Pareto-optimal ($\star$) if no other condition achieves both greater suppression *and* higher coherence retention simultaneously. All values are computed on writing prompt evaluations ($n = 20$ per condition for CAA runs; $n = 5$ per window for the layer sweep).

1. Standard CAA and Detector-Weighted CAA

Table 12: Pareto calculations for standard CAA and DW-CAA variants on writing prompts — QWEN2.5-3B. Baseline: AI_0=6.20, Coh_0=3.15. $\star$ = Pareto-optimal.

Run	Condition	AI_i	Coh_i	Suppression	Coh. retention
Standard CAA	Best-layer	5.90	3.05	0.30	0.968 $\star$
Standard CAA	All-layer	2.10	1.80	4.10	0.571 $\star$
DW-RoBERTa	Best-layer	6.05	3.20	0.10	0.985 $\star$
DW-RoBERTa	All-layer	3.65	2.00	2.50	0.615 $\star$
DW-RADAR	Best-layer	6.00	3.20	0.20	0.941
DW-RADAR	All-layer	2.70	1.65	3.50	0.485
DW-GPTZero	Best-layer	5.85	2.80	0.30	0.875
DW-GPTZero	All-layer	2.85	1.80	3.30	0.562

Table 13: Pareto calculations for standard CAA and DW-CAA variants on writing prompts — QWEN2.5-3B-INSTRUCT. Baseline: AI_0=6.15, Coh_0=3.60. $\star$ = Pareto-optimal.

Run	Condition	AI_i	Coh_i	Suppression	Coh. retention
Standard CAA	Best-layer	5.80	3.40	0.35	0.944 $\star$
Standard CAA	All-layer	2.15	2.50	4.00	0.694 $\star$
DW-RoBERTa	Best-layer	5.90	3.50	0.20	0.959 $\star$
DW-RoBERTa	All-layer	2.50	2.65	3.60	0.726
DW-RADAR	Best-layer	6.10	3.40	0.15	1.015 $\star$
DW-RADAR	All-layer	2.65	2.45	3.60	0.731
DW-GPTZero	Best-layer	5.65	3.15	0.45	0.900 $\star$
DW-GPTZero	All-layer	2.15	2.85	3.95	0.814 $\star$

2. Layer Sweep Pareto-Optimal Windows

Tables 14–17 report the full Pareto calculations for the layer sweep across all four conditions (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT) $\times$ ($k = 1$, $k = 5$). Only Pareto-optimal windows are marked; all windows are included for completeness. Reference conditions are given in each caption.

Table 14: Layer sweep Pareto calculations — QWEN2.5-3B, $k = 1$. Baseline: AI_0=6.20, Coh_0=3.35. All-layer reference: x=4.05, y=0.567. $\star$ = Pareto-optimal.

Window	Center	AI_i	Coh_i	Suppression	Coh. retention
0	0	6.20	3.40	0.00	1.015
1	1	6.00	3.00	0.20	0.896
2	2	6.00	3.80	0.20	1.134 $\star$
3	3	6.00	3.40	0.20	1.015
4	4	6.60	3.40	−0.40	1.015
5	5	6.20	3.60	0.00	1.075
6	6	6.00	2.80	0.20	0.836
7	7	6.20	3.00	0.00	0.896
8	8	6.00	2.80	0.20	0.836
9	9	6.20	3.60	0.00	1.075
10	10	6.00	3.20	0.20	0.955
11	11	6.00	3.20	0.20	0.955
12	12	6.20	2.80	0.00	0.836
13	13	6.00	3.00	0.20	0.896
14	14	6.20	2.60	0.00	0.776
15	15	6.20	3.20	0.00	0.955
16	16	6.00	3.00	0.20	0.896
17	17	5.20	2.60	1.00	0.776
18	18	5.40	2.40	0.80	0.716
19	19	4.60	2.00	1.60	0.597
20	20	6.00	2.60	0.20	0.776
21	21	5.60	2.20	0.60	0.657
22	22	5.00	2.00	1.20	0.597
23	23	5.40	2.80	0.80	0.836
24	24	5.20	2.00	1.00	0.597
25	25	5.00	2.20	1.20	0.657
26	26	5.00	2.80	1.20	0.836
27	27	5.80	3.00	0.40	0.896
28	28	5.80	3.20	0.40	0.955 $\star$
29	29	6.00	3.60	0.20	1.075
30	30	5.60	2.40	0.60	0.716
31	31	6.20	3.00	0.00	0.896
32	32	5.20	3.20	1.00	0.955 $\star$
33	33	4.60	3.00	1.60	0.896 $\star$
34	34	5.80	3.00	0.40	0.896
35	35	5.80	3.00	0.40	0.896

Table 15: Layer sweep Pareto calculations — QWEN2.5-3B, $k = 5$. Baseline: AI_0=6.20, Coh_0=3.20. All-layer reference: x=4.15, y=0.500. $\star$ = Pareto-optimal.

Window	Center	AI_i	Coh_i	Suppression	Coh. retention
0	2	6.20	3.60	0.00	1.125
1	3	6.00	3.00	0.20	0.938
2	4	6.40	3.60	−0.20	1.125
3	5	6.00	3.60	0.20	1.125 $\star$
4	6	6.20	3.80	0.00	1.187 $\star$
5	7	6.40	3.20	−0.20	1.000
6	8	5.20	2.60	1.00	0.812
7	9	6.00	3.00	0.20	0.938
8	10	5.80	2.80	0.40	0.875
9	11	6.20	3.80	0.00	1.187 $\star$
10	12	5.80	2.40	0.40	0.750
11	13	6.40	2.80	−0.20	0.875
12	14	5.40	3.60	0.80	1.125 $\star$
13	15	5.60	2.80	0.60	0.875
14	16	4.60	3.00	1.60	0.938 $\star$
15	17	5.40	3.00	0.80	0.938
16	18	4.60	2.80	1.60	0.875
17	19	5.00	2.20	1.20	0.688
18	20	5.20	1.80	1.00	0.562
19	21	5.00	1.80	1.20	0.562
20	22	5.40	2.00	0.80	0.625
21	23	3.80	2.20	2.40	0.688 $\star$
22	24	4.60	1.60	1.60	0.500
23	25	3.40	2.00	2.80	0.625 $\star$
24	26	5.20	2.60	1.00	0.812
25	27	4.60	2.80	1.60	0.875
26	28	4.80	2.40	1.40	0.750
27	29	5.20	2.20	1.00	0.688
28	30	5.00	3.00	1.20	0.938 $\star$
29	31	4.80	3.00	1.40	0.938 $\star$
30	32	5.00	2.20	1.20	0.688
31	33	5.20	3.00	1.00	0.938 $\star$

Table 16: Layer sweep Pareto calculations — QWEN2.5-3B-INSTRUCT, $k = 1$. Baseline: AI_0=6.05, Coh_0=3.55. All-layer reference: x=3.40, y=0.690. $\star$ = Pareto-optimal.

Window	Center	AI_i	Coh_i	Suppression	Coh. retention
0	0	5.80	3.40	0.25	0.958
1	1	6.00	3.20	0.05	0.901
2	2	6.00	3.20	0.05	0.901
3	3	5.80	3.20	0.25	0.901
4	4	5.60	3.80	0.45	1.070 $\star$
5	5	6.00	3.80	0.05	1.070
6	6	6.00	3.20	0.05	0.901
7	7	5.80	3.40	0.25	0.958
8	8	5.60	3.40	0.45	0.958
9	9	6.00	3.80	0.05	1.070
10	10	5.80	3.40	0.25	0.958
11	11	6.00	3.40	0.05	0.958
12	12	6.00	3.20	0.05	0.901
13	13	6.00	4.00	0.05	1.127 $\star$
14	14	5.60	3.20	0.45	0.901
15	15	6.00	3.80	0.05	1.070
16	16	6.20	3.40	−0.15	0.958
17	17	5.20	3.20	0.85	0.901
18	18	5.40	2.80	0.65	0.789
19	19	4.20	2.80	1.85	0.789 $\star$
20	20	5.20	3.20	0.85	0.901
21	21	5.00	2.80	1.05	0.789
22	22	4.40	3.40	1.65	0.958 $\star$
23	23	5.40	2.80	0.65	0.789
24	24	4.80	3.00	1.25	0.845
25	25	5.00	3.40	1.05	0.958
26	26	4.80	3.60	1.25	1.014 $\star$
27	27	5.80	3.60	0.25	1.014
28	28	5.40	3.20	0.65	0.901
29	29	6.00	3.40	0.05	0.958
30	30	5.80	3.20	0.25	0.901
31	31	6.00	3.00	0.05	0.845
32	32	5.00	3.20	1.05	0.901
33	33	5.40	3.00	0.65	0.845
34	34	5.80	3.00	0.25	0.845
35	35	6.20	3.60	−0.15	1.014

Table 17: Layer sweep Pareto calculations — QWEN2.5-3B-INSTRUCT, $k = 5$. Baseline: AI_0=6.10, Coh_0=3.50. All-layer reference: x=3.40, y=0.700. $\star$ = Pareto-optimal.

Window	Center	AI_i	Coh_i	Suppression	Coh. retention
0	2	5.80	3.80	0.30	1.086 $\star$
1	3	5.80	3.20	0.30	0.914
2	4	6.00	3.60	0.10	1.029
3	5	6.00	3.40	0.10	0.971
4	6	6.20	3.60	−0.10	1.029
5	7	6.00	3.20	0.10	0.914
6	8	5.80	3.40	0.30	0.971
7	9	6.20	3.60	−0.10	1.029
8	10	5.80	2.80	0.30	0.800
9	11	5.40	3.40	0.70	0.971
10	12	6.00	3.00	0.10	0.857
11	13	5.80	3.40	0.30	0.971
12	14	5.80	3.00	0.30	0.857
13	15	5.20	2.80	0.90	0.800
14	16	5.60	3.40	0.50	0.971
15	17	5.80	3.00	0.30	0.857
16	18	3.20	2.80	2.90	0.800 $\star$
17	19	4.20	2.40	1.90	0.686
18	20	4.20	3.60	1.90	1.029 $\star$
19	21	4.40	3.20	1.70	0.914
20	22	4.80	2.60	1.30	0.743
21	23	4.40	2.60	1.70	0.743
22	24	4.40	3.00	1.70	0.857
23	25	4.40	2.80	1.70	0.800
24	26	4.60	2.80	1.50	0.800
25	27	3.80	2.60	2.30	0.743
26	28	4.40	2.80	1.70	0.800
27	29	5.80	3.60	0.30	1.029
28	30	4.00	2.60	2.10	0.743
29	31	5.60	2.60	0.50	0.743
30	32	4.00	2.80	2.10	0.800 $\star$
31	33	5.40	3.00	0.70	0.857

K Capability Benchmarks: Full Results

Tables 18 and 19 report absolute scores and deltas for all four capability benchmarks across all eight runs and three ablation conditions. MMLU and HellaSwag are reported as accuracy (0–1); TruthfulQA is MC1 accuracy (0–1); perplexity is WikiText-2 perplexity (lower is better). Deltas are from baseline: pp for accuracy benchmarks, absolute for perplexity.

Table 18: Capability benchmarks — base models (QWEN2.5-3B). Absolute scores and deltas from baseline. PPL = WikiText-2 perplexity.

		MMLU		HellaSwag		TruthfulQA		PPL	
Run	Condition	Score	Δ (pp)	Score	Δ (pp)	Score	Δ (pp)	Score	Δ
Std. CAA	Baseline	0.694	—	0.645	—	0.620	—	126.6	—
	Best-layer	0.696	+0.2	0.625	−2.0	0.605	−1.5	143.4	+16.7
	All-layer	0.702	+0.8	0.620	−2.5	0.540	−8.0	137.3	+10.6
DW-RoBERTa	Baseline	0.694	—	0.650	—	0.620	—	126.6	—
	Best-layer	0.688	−0.6	0.625	−2.5	0.605	−1.5	137.5	+10.9
	All-layer	0.688	−0.6	0.645	−0.5	0.530	−9.0	132.3	+5.6
DW-RADAR	Baseline	0.694	—	0.650	—	0.620	—	126.6	—
	Best-layer	0.690	−0.4	0.625	−2.5	0.605	−1.5	136.5	+9.9
	All-layer	0.684	−1.0	0.630	−2.0	0.535	−8.5	128.2	+1.5
DW-GPTZero	Baseline	0.694	—	0.645	—	0.620	—	126.6	—
	Best-layer	0.696	+0.2	0.630	−1.5	0.605	−1.5	139.8	+13.2
	All-layer	0.696	+0.2	0.620	−2.5	0.530	−9.0	148.7	+22.1

Table 19: Capability benchmarks — instruct models (QWEN2.5-3B-INSTRUCT). Absolute scores and deltas from baseline.

		MMLU		HellaSwag		TruthfulQA		PPL	
Run	Condition	Score	Δ (pp)	Score	Δ (pp)	Score	Δ (pp)	Score	Δ
Std. CAA	Baseline	0.704	—	0.665	—	0.720	—	474.7	—
	Best-layer	0.706	+0.2	0.645	−2.0	0.715	−0.5	518.3	+43.6
	All-layer	0.700	−0.4	0.655	−1.0	0.670	−5.0	430.9	−43.8
DW-RoBERTa	Baseline	0.704	—	0.665	—	0.720	—	474.7	—
	Best-layer	0.706	+0.2	0.645	−2.0	0.715	−0.5	536.0	+61.4
	All-layer	0.700	−0.4	0.655	−1.0	0.660	−6.0	440.7	−34.0
DW-RADAR	Baseline	0.704	—	0.665	—	0.720	—	474.7	—
	Best-layer	0.706	+0.2	0.630	−3.5	0.720	0.0	535.5	+60.8
	All-layer	0.704	0.0	0.645	−2.0	0.715	−0.5	424.1	−50.6
DW-GPTZero	Baseline	0.704	—	0.665	—	0.720	—	474.7	—
	Best-layer	0.708	+0.4	0.640	−2.5	0.715	−0.5	508.9	+34.2
	All-layer	0.708	+0.4	0.650	−1.5	0.665	−5.5	402.4	−72.3

L Vocabulary Signatures via Logit Lens

To better understand what the extracted AI-style direction encodes, we project it onto the model's unembedding matrix W_U to produce a vocabulary-space signature $\mathbf{v} = W_U\hat{\mathbf{d}}$. Tokens with the largest positive entries are those whose generation probability would increase most if the direction were added to a residual stream hidden state. Tokens with the most negative entries form the human-style vocabulary signature. Results are broadly consistent across detector variants and model types.

Three interpretable clusters emerge in the AI-style tokens: emojis and markdown formatting, formal text, and web code and artifacts. First, emojis and markdown formatting tokens are characteristic of structured AI-generated content, where visual emphasis and bullet-point formatting are common. Second, Japanese formal constructions (ことができます, することができます, meaning roughly "is able to" and "can do" in polite register) likely reflect Qwen's multilingual pretraining, in which formal Japanese text is distributed similarly to formal English AI outputs in the residual stream. Third, web and code artifacts (`creativecommons`, ®,) suggest the direction also picks up on structured reference and attribution conventions common in AI-generated explanatory text.

The human-style tokens form a coherent contrasting cluster of almost entirely informal English consisting of profanity, hedging qualifiers (`maybe`, `probably`, `apparently`), and emphatic colloquialisms (`utterly`, `totally`). This is consistent with the HC3 human responses originating predominantly from the ELI5 subreddit, where casual, opinionated, and colloquial language is the norm. The direction thus accurately captures the stylistic divergence between informal human discourse and polished AI outputs at the vocabulary level.